

Astronomical Techniques - Astronomical Photography

Photography was for decades the only objective recording technique available to astronomers, with advantages in area covered, ability to integrate light over long times, and archival longevity. It has been overtaken in sensitivity and accuracy by CCDs, which are finally approaching the ability of photographic plates to cover large areas at once. Why should we still deal with photography despite its low quantum efficiency (seldom greater than 1%)?

There are several rationales. There is still a century-long archive of photographic data to understand and use. Photography had, until quite recently, the edge in wide-field coverage (mosaics of the largest CCD chips have grown from 6 to >25 cm by now), and was the only practical means of doing optical sky surveys. The ease of storage of photographic data has much to recommend it, and it is incomparably cheap and easy to use without support equipment (the UIT shuttle experiment used photography so that no expendable supplies ran out during delays on the launch pad). Finally, photography may still be the last step in reporting even digital data, so familiarity with the process has its uses.

PHOTOGRAPHIC CHEMISTRY

The chemistry of photography lies somewhere between solid-state physics and black magic. Certain silver compounds (especially with halides, elements from column 7a of the periodic table such as bromine, chlorine, and iodine) have crystal structures such that absorption of a visible-light photon liberates an electron into a valence band, creating an electron-hole pair each of which can migrate freely. If certain impurities exist that can trap the electron (typically by neutralizing a silver ion), there can be a long-lasting state change. These traps can actually attract more free electrons. During development, the emulsion is exposed to a solution which slowly transforms the silver halides to silver. The gain of photographic detection comes in with the fact that silver atoms are strong catalysts for this process, with a gain approaching 10^9 . Finally, a fixing step removes excess silver atoms and renders the image permanent. Additional steps to remove remaining developer and refixing can yield an image capable of lasting more than 100 years.

The silver halide is clumped in grains, whose size determines the sensitivity and resolution of the process. These are imbedded in a gelatin layer (forming the emulsion) deposited on a glass or plastic base (plates or film). The final image is negative, more opaque where the most light was detected.

The speed of an emulsion measures its sensitivity, being gauged by the time taken to reach a particular opacity for a given source. ISO ratings (formerly ASA numbers) are linear, DIN (Deutsche Industrie-Norm) are logarithmic (see textbook p. 195). All other things being equal, larger grains will detect light more efficiently, but since the resolution is limited to the typical grain size, there is a tradeoff. Chemists have learned to make the grains flat, so that there can be maximum surface area for detection at a given resolution.

The amount of detected light is measured by the photographic density, which can be found by shining a narrow beam of light through the emulsion at a desired point and comparing the intensity F_{in} before and F_{out} after passing through the photograph; the density is defined by $D = \log (F_{in}/F_{out})$. Note that there can be instrumental subtleties in making sure that only the light going straight through the plate is measured; scattered light can be a major problem for high densities. The transmission is then 10^{-D} and the opacity is 10^D , both now in linear units. The relation between density and incident intensity is nonlinear and changes for different batches of the same emulsion as well as depending on processing details. Its form is given by the HD (Hurter-Driffield) or characteristic curve (textbook p. 199), consisting of an initial density (the fog level) independent of light, a slow ramping upwards (the toe), a linear portion, and finally a ramping into another flat portion at saturation. This

curve is conventionally displayed as log density versus log exposure. The slope of the linear portion is the contrast of the emulsion, which depends on its composition and development. Different applications call for various levels of contrast, since this governs the latitude of the emulsion - how much intensity range can be recorded in a single exposure.

Internal calibration sources and step wedges are used to determine at least a relative calibration curve for astronomical photographs. The step wedge contains glass with a set of regions (usually boxes) of known photographic density, and thus, whatever source shines through them, known relative intensity. A light source is usually set so dim that it can be on during the entire astronomical exposure, to replicate the observing conditions as completely as possible. Some kinds of images can also calibrate themselves, such as using a rich star field and the requirement that the intensity profiles of all the stars must be the same, or in a spectrogram when we know that two emission lines must have a certain fixed intensity ratio and the same profiles.

Different emulsions have different spectral response, which are frequently modified further by use of filters. Raw silver halides are most sensitive in the blue and near-ultraviolet, but various dyes can bond chemically to the grains so as to modify the sensitivity (as far as almost 1 micron in some cases). For spectroscopy, one wants a smooth variation of response with wavelength, while for direct imaging one wants the highest overall sensitivity within a given passband. Color emulsions (almost never used at the telescope except for publicity or planetary work) use three sets of differently responsive emulsions sandwiched together.

Filters are usually glass, either impregnated with various dyes for broadband filters or coated with thin films for narrow-band interference filters. One can use a filter to define a predetermined passband to ease standardization, to isolate a spectral range or component of special interest, or block unwanted light (for example, to get rid of the red light of a galaxy so as to see faint blue clusters of younger stars). Filters need to be adequately blocked - that is, have zero response to unwanted light across the entire sensitivity band of the detector. A common problem with blue and UV filters is a red leak - a few percent leak well to the red of the nominal passband. This can be a real gotcha for UV observations of old galaxies, which are 50-100 times fainter in the UV than in visible light, and the bandwidth ratio also works the wrong way. There have been some problems as well with interference filters that worked adequately with photographic plates and TV detectors, but have red leaks from 0.8-1.0 micron, where many CCDs are sensitive.

The emulsions most commonly used in astronomy were long the so-called spectroscopic emulsions from Kodak, such as 103a-O, IIa-O, IIIa-J, and IIIa-F. The "a" denotes a special process to reduce reciprocity failure, the greatest special problem in astronomical photography. Normally, the product of intensity and exposure time gives the level of opacity in the emulsion (the reciprocity law). However, for long exposures, the emulsion becomes relatively less efficient, so that for ordinary emulsions long exposures eventually give no added advantage. Astronomical emulsions are designed to minimize this, so that while slower for very short exposures, they are much faster at low light levels and long exposures. There also exist various sensitizing or hypersensitizing techniques that may decrease reciprocity failure and fog as well as increasing sensitivity. These include:

- cooling, often to solid CO₂ temperatures (dry ice). This reduces thermal effects, and was one of the first effective ways of reducing reciprocity failure.
- baking at a few hundred degrees C in either nitrogen, hydrogen (very dangerous!) or forming gas (nitrogen with 5% hydrogen). Even vacuum baking often helps. Gains from this step are so large that class III emulsions are almost never used without such preprocessing.
- liquid baths, removing silver halide ions to increase the concentration of silver ions.
- preflashing, applying enough extra exposure to put the emulsion above the toe of its HD curve. This is not so much sensitizing as making most effective use of the emulsion's existing properties for detecting faint sources.

There are artifacts to which photography is subject, some apparent only when trying the level of quantitative analysis needed for some astronomical applications.

- Saturation. The emulsion cannot record any information above the exposure level corresponding to D_{max} , and the information transfer dies off rapidly as this is approached. Thus there is a limit to the dynamic range of information on a single photograph. For exposure levels far above this, the density may actually decrease slightly (solarization).
- Halation. Light may scatter off the back of the film or plate, and be detected when passing backwards through the emulsion. This gives scattered halos around stars, and can be controlled with an absorbing coating on the back, removed during processing.
- Exhaustion. In a dense region, the developer may become diluted and less able to function. Thus, for example, the area between two close stars will look too light (less dense on the negative), and edges may be enhanced.

Processing can be very exacting if quantitative results are needed. Time and temperature must be closely controlled, and fresh chemicals should be used. A few emulsions can tolerate safelights, but it is good practice to process in total darkness. Cleanness is a must to avoid dust specks on the emulsion, especially before processing where they can leave permanent spots. Agitation is needed to remove air bubbles and redistribute chemicals during development. For archival longevity, the washing and fixing steps may be broken into two baths so the final ones have minimal residual contamination from the earlier steps.

Contrast can be somewhat modified by development time; longer development gives higher contrast but higher chemical fog as well. One should strive for uniformity in any particular project, to get repeatable results. Emulsions should not be touched, as they are very fragile. Handle only the edges or (when unavoidable) the backs.

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